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## **IT-7005(3)-CBGS**

### **B.E. VII Semester**

Examination, June 2020

## **Choice Based Grading System (CBGS)**

### **Embedded System**

**Time : Three Hours**

**Maximum Marks : 70**

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. a) What is an Embedded system? Why is it so hard to define?  
b) List and define three main characteristics of embedded systems that distinguish such systems from other computing systems.
2. a) Define the use of Memory, Sensors and Actuators in embedded systems.  
b) What are the fundamental issues are there in hardware software co-design?
3. a) What is Finite State Machine (FSM) model of embedded system? Explain with suitable figure.  
b) Draw and explain concurrent model. Differentiate it with other models.
4. a) Give an architectural view of 8085 microcontroller.  
b) List various registers of 8085 and explain the purpose of each one of them.

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5.
  - a) Give instruction set completeness of 8085. Also write addressing modes.
  - b) Differentiate OS based and super loop based embedded firmware design approaches.
6.
  - a) Give an overview of assembly language based and high level language based firmware development languages.
  - b) What is Operating System? What is the use of it?
7.
  - a) Differentiate multiprocessing and multitasking operating system.
  - b) What do you know about task scheduling? Write different methods of task, scheduling.
8. Write short notes on any two of the following.
  - a) Task synchronization
  - b) Interrupts of 8085
  - c) Programming in Embedded C

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